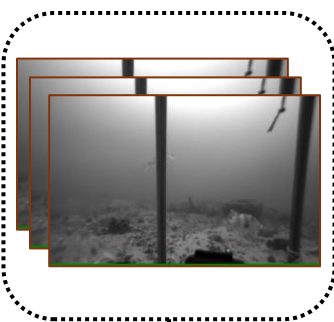




(1) Object Detection (YOLOX)

High confidence
> 0.5

Low confidence
0.01-0.5

(2) IMM Motion Predictor

KF_{CV}

(Constant Velocity)
Straight Swimming

KF_{CA}

(Constant Acceleration)
Speed-up / Slow-down

Tracklets
 T_{k-1}

Predicted IMM Tracks
 T_k

(3) Association (FishIoU + Cost + Hungarian)

Stage 1 Association (High conf. detections + Predicted tracks)

FishIoU

- IoU
- Aspect ration consistency
- Area consistency
- Center distance penalty

Cost Matrix
(1- FishIoU)

Hungarian Association

Matched Tracks

Unmatched Detections

Unmatched Tracks

New Tracks

Stage 2 Association (Low conf. detections + Unmatched tracks from stage1)

FishIoU

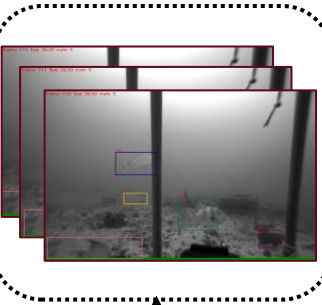
- IoU
- Aspect ration consistency
- Area consistency
- Center distance penalty

Cost Matrix
(1- FishIoU)

Hungarian Association

Matched Tracks

Unmatched Tracks



Lost Tracks

Updated Track T_k